1 Contrapositive

1. $P \implies Q = \sim Q \implies \sim P$ and outline of the proof.
2. Suppose $x \in \mathbb{R}$. If $x^2 + 5x < 0$, then $x < 0$.
3. When to do contrapositive
4. Prove "If $5x + 7$ is even, then x is odd" directly and by contrapositive.
5. Prove "If $x < y$, then $x^3 < y^3$ directly and by contrapositive.
6. Suppose $x, y \in \mathbb{Z}$. If $7 \nmid xy$, then $7 \nmid x$ and $7 \nmid y$.
7. For $n \in \mathbb{N}$, if $2^n - 1$ is prime, then n is prime.

2 Modular congruence

1. a and b are **congruent modulo** n, $a \equiv b \pmod{n}$ iff $n \mid (b - a)$; $a \not\equiv b$
2. random examples
3. $a \equiv b \pmod{n} \iff a = qn + r, b = q'n + r', r = r'$
4. $a, b \in \mathbb{Z}$, $n \in \mathbb{N}$. $a \equiv b \pmod{n} \implies a^2 \equiv b \pmod{n}$.
5. $a \equiv b \pmod{n} \implies ca \equiv cb \pmod{n}$
6. $ac \not\equiv bc \pmod{m} \implies n \nmid c$

3 Contradiction

1. $P =$ "$(\sim P) \implies (C \wedge \sim C)''$, and outline of the proof.
2. If $a, b \in \mathbb{Z}$, then $a^2 - 4b \neq 2$.
3. Define rational/irrational number.
4. $\sqrt{2}$ is irrational
5. There are infinitely many prime numbers
6. The function $f(x) = x^3 + x - 2$ has only one root.

4 Proving $P \implies Q$ by contradiction

1. Outline of the proof
2. $a \in \mathbb{Z}$. If a^2 is even, then a is even.
3. If $a, b \in \mathbb{Z}$ and $a \geq 2$, then $a \nmid b$ or $a \nmid (b + 1)$.